

## 2-Mark Questions & Answers MPCA Module 6: Pentium 4 & ARM Processor

---

### Q1. What is NetBurst microarchitecture in Pentium 4?

The **NetBurst Microarchitecture** is Intel's 7th-gen x86 design featuring a **20-stage hyper-pipeline**, Execution Trace Cache, and Rapid Execution Engine to achieve high clock speeds (up to 3.8 GHz).

---

### Q2. What is Execution Trace Cache in Pentium 4?

The **Execution Trace Cache** replaces the traditional L1 instruction cache by storing up to **12,000 pre-decoded micro-operations (uops)** in their predicted execution sequence, bypassing the instruction decoder.

---

### Q3. Define Hyper-Threading Technology (HT).

**Hyper-Threading** is Simultaneous Multithreading (SMT) that allows a single physical processor core to function as **two logical processors** by duplicating architectural registers while sharing execution units.

---

### Q4. What is the Inline Barrel Shifter in ARM processors?

The **Inline Barrel Shifter** is an on-chip hardware unit in ARM that can shift or rotate one operand by arbitrary bit positions within the **same single clock cycle** as arithmetic/logical operations.